



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Class : CSE

Batch : 2025-29

Assignment :

Subject: Programming with Java

ASSIGNMENT No : 8

TITLE : Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. Super helps distinguish between parent and child class members with the same name. It ensures proper constructor chaining during object creation. It is an important concept in inheritance and method overriding.



Code and Output Screenshots

```
J Main5.java > Java > Main5
1  class Parent {
2      void display () {
3          System.out.println("This is a parent class method");
4      }
5  }
6
7  class Child extends Parent {
8      void display() {
9          super.display();
10         System.out.println("This is a child class method");
11     }
12 }
13
14 public class Main5 {
15     Run main | Debug main | Run | Debug
16     public static void main (String[] args) {
17         Child c = new Child();
18         c.display();
19     }
20 }
```

PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS Filter (e.g. text, !excludeText, t...

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main5.java && java Main5
This is a parent class method
This is a child class method

[Done] exited with code=0 in 0.508 seconds
```

Conclusion

This program demonstrates method overriding in Java. The child class overrides the parent class method and uses `super.display()` to call the parent class method before executing its own. This shows how inheritance and method overriding help extend and customize the behavior of existing classes.



Exercise

1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

```
J Exr1.java > Java > Exr1
1  class Ex1 {
2      void display() {
3          System.out.println("This class is employee");
4      }
5  }
6
7  class Manager extends Ex1{
8      void display() {
9          super.display();
10         System.out.println("This class is manager");
11     }
12 }
13
14 public class Exr1 {
15     Run main | Debug main | Run | Debug
16     public static void main (String[] args) {
17         Manager m = new Manager();
18         m.display();
19     }
```

PROBLEMS 8 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, !excludeText, t...

```
[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Exr1.java && java Exr1
This class is employee
This class is manager
```

```
[Done] exited with code=0 in 0.47 seconds
```



2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

```
J Main6.java > Language Support for Java(TM) by Red Hat > Main6
1  class Vehiclex2 {
2      String vname;
3
4      Vehiclex2(String vname) {
5          this.vname = vname;
6      }
7      void display() {
8          System.out.println("Vehicle name: " + vname);
9      }
10 }
11 class Insurance extends Vehiclex2 {
12     String insuranceCompany;
13     Insurance(String vname, String insuranceCompany) {
14         super(vname);
15         this.insuranceCompany = insuranceCompany;
16     }
17     void display() {
18         super.display();
19         System.out.println("Insurance Company: " + insuranceCompany);
20     }
21 }
22 public class Main6 {
23     Run main | Debug main | Run | Debug
24     public static void main(String[] args) {
25         Insurance insurance = new Insurance("Car", "Acko Insurance");
26         insurance.display();
27     }
28 }
```

PROBLEMS 12 OUTPUT DEBUG CONSOLE TERMINAL PORTS

Filter (e.g. text, !excludeText, t...

[Running] cd "c:\Users\LENOVO\Documents\JAVA Assignments\" && javac Main6.java && java Main6

Vehicle name: Car

Insurance Company: Acko Insurance

[Done] exited with code=0 in 0.528 seconds